

Pronounceable Words (words)

William loves words. Today he is trying to come up with “pronounceable” words. What is a pronounceable word? Well, clearly “hello” is a pronounceable word, while “tqwukjt” is not. William defines a word as *pronounceable* if every one of its 3-character substrings is an *acceptable trigram*.

In the example above, the “wuk” substring could be reasonably pronounced so it’s indeed an acceptable trigram, “qwu” is a bit ugly but it could still pass, but then what about “tqw” and “kjt”? They are just too weird to say!



To make things easy, William collected a list of all the K acceptable trigrams. With the hard part out of the way, now comes the easy stuff: generating the pronounceable words.

William is asking for your help. He left a note where he detailed how he wants you to write a program that reads in the *known trigrams* and two numbers N and L , and then performs the following operations:

1. Calculate a list of all the pronounceable words that are exactly L -characters long.
2. Sort the list alphabetically.
3. Finally, print the N -th word from the list.

Consider that there might be *a lot* of pronounceable words, and you’re only required to print one word... It’s OK if you don’t actually compute all the words, no one needs to know.

 Among the attachments of this task you may find a template file `words.*` with a sample incomplete implementation.

Input

The first line contains three integers: N , L , and K . Each of the next K lines contains a 3-character string S_i .

Output

You need to write a single line with a L -letter long word: the N -th “pronounceable” word.

Constraints

- $1 \leq N \leq 10^{18}$.
- There are at least N pronounceable words.
- $3 \leq L \leq 10\,000$.
- $1 \leq K \leq 17\,576 = 26^3$.
- The length of S_i is exactly 3 characters, for each $i = 0 \dots K - 1$.
- S_i is always formed by lowercase alphabetic letters, and all trigrams are distinct.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (10 points) $N = 1, L \leq 10$.

- **Subtask 3** (30 points) $N \leq 20, L \leq 10$.

- **Subtask 4** (30 points) $N \leq 1000$.

- **Subtask 5** (30 points) No additional limitations.


Examples

input	output
3 4 4 aba ada dab bad	bada
52 10 5 aaa baa aba abb aab	baaaaaaab

Explanation

In the **first sample case** there are exactly 4 pronounceable words that are 4 characters long: **abad**, **adab**, **bada**, **daba**.

In the **second sample case** there are exactly 73 pronounceable words that are 10 characters long.